Summer Examinations 2018-19

Exam Code(s)	3BA1, 4BA1, 4BME1, 1OA1, 3BMS2, 3BS91, 3FM2, 1HA1
Exam	Third & Fourth Year & HDip
Module	TOPOLOGY
Module Codes	MA342 & MA535
External Examiner(s)	Prof T. Brady
Internal Examiner(s)	Prof G. Ellis*
Instructions	Attempt **all** questions. There are ten questions, each carrying 10 marks.
Duration	2 hours
No. of Pages	3 pages (including this cover page)
Discipline	Mathematics

Requirements:

Release in Exam Venue	Yes ☑	No ☐
Release to Library	Yes ☑	No ☐
Non-programmable calculator	Yes ☐	No ☑
Mathematical Tables	Yes ☐	No ☑

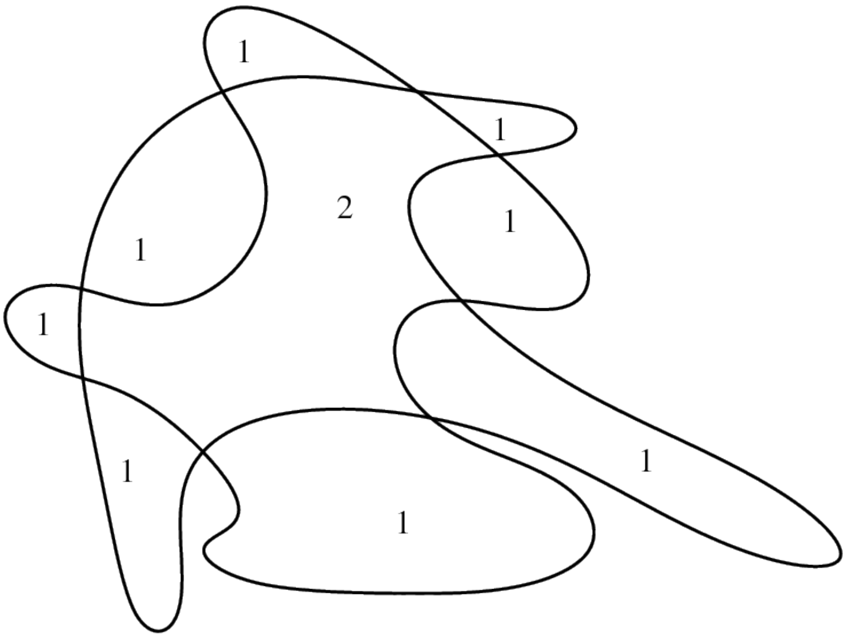
1. Explain what is meant by a triangulation of a topological space. Exhibit a triangulation of the torus

$$T = [0,1] \times [0,1]/\{(x,0) = (x,1), (0,y) = (1,y) \text{ for } 0 \leq x, y \leq 1\}$$

and use it to calculate the Euler characteristic $\chi(T)$.

2. Suppose that a graph Γ with V vertices and E edges is embedded in the 2-dimensional sphere $\mathbb{S}^2$ in such a way that edges intersect only at vertices and that the complement $\mathbb{S}^2 \setminus \Gamma$ is a disjoint union of F homeomorphic copies of the open unit 2-dimensional disk. Prove that $V - E + F = 2$.

3. The following picture shows the boundaries of several closed subspaces $U_1, \ldots, U_t \subseteq \mathbb{E}^2$ of common Euler characteristic $\chi(U_i) = 1$. No two boundaries are tangential.



Let $X = U_1 \cup \ldots \cup U_t$. The numbers in the interiors of the subspaces and their intersections represent the weight function $\omega\colon X \to \mathbb{Z}$ where $\omega(x) = |\{i : x \in U_i\}|$. Define the Euler integral

$$\int_X \omega \, d\chi$$

and then calculate this integral, showing your calculations. Deduce the number of subspaces t from your calculation.

4. For each of the following sets X and collections T of open subsets decide whether the pair X, T satisfies the axioms of a topological space. If it does, determine the connected components of X. If it is not a topological space then exhibit one axiom that fails.

(a) $X = \{1,2,3,4\}$ and $T = \{\emptyset, \{1\}, \{1,2\}, \{2,3\}, \{1,2,3\}, \{1,2,3,4\}\}$.
(b) $X = \{1,2,3,4\}$ and $T = \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1,2,3\}, \{1,2,3,4\}\}$.
(c) $X = \mathbb{R}$ and a subset $U \subset \mathbb{R}$ is open if and only if U is infinite or $U = \emptyset$.
(d) $X = \mathbb{R}$ and a subset $U \subset \mathbb{R}$ is open if and only if $\mathbb{R} \setminus U$ is countable. (A set is *countable* if, and only if, it is finite or in bijection with $\mathbb{Z}$.)

5. Let the set $\mathbb{N}$ of natural numbers be endowed with the cofinite topology (in which a set is open if and only if it is empty or its complement is finite).

(a) Is $\mathbb{N}$ connected? Justify your answer.
(b) Is $\mathbb{N}$ compact? Justify your answer.
(c) Explain why the function $f\colon \mathbb{N} \to \mathbb{N}, n \mapsto n^3$ is continuous.
(d) Exhibit a function $g\colon \mathbb{N} \to \mathbb{N}$ which is not continuous.

6. Answer **either** (a) **or** (b).

(a) Prove that a subset A of a topological space X is *closed* if and only if it contains all its *accumulation points*. Include definitions of the two italicized terms.
(b) Prove that *compactness* is a *topological property*. Include definitions of the two italicized terms.

7. Prove that the circle $\mathbb{S}^1 = \{(x, y) \in \mathbb{E}^2 : x^2 + y^2 = 1\}$ is not homeomorphic to the unit interval $[0, 1]$. State clearly any theorem that you use.

8. Prove that the circle $\mathbb{S}^1$ is *homotopy equivalent* to the punctured complex plane $\mathbb{C}\setminus\{0\}$. Include a definition of the italicized term.

9. Answer **either** (a) **or** (b) **or** (c).

(a) State the Brouwer Fixed-Point Theorem and use it prove that any square matrix with positive real entries has a positive real eigenvalue with corresponding eigenvector having only non-negative real entries.
(b) Prove that the unit interval $[0, 1]$ is compact.
(c) Prove that any polynomial $a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$ with coefficients $a_i \in \mathbb{C}$ and degree $n > 0$ has at least one zero in $\mathbb{C}$. You may use the bijection $[S^1, S^1] \cong \mathbb{Z}$ that associates the homotopy class of a map with the winding number of the map.

10. Answer **either** (a) **or** (b) **or** (c).

(a) Prove that a compact subset of a Haussdorff topological space is closed.
(b) Use the fact that the Euler characteristic of a triangulable space is a homotopy invariant to prove Brouwer's fixed point theorem.
(c) Describe what is meant by a *Nash Equilibrium*, explaining any concepts from Game Theory that you use.